# Ch 6.2: Shrinkage - The Lasso

## Lecture 18 - CMSE 381

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Mon, Oct 15, 2024

# Announcements

**Last time:**

- Ridge Regression

**This time:**

- The Lasso

**Announcements:**

- HW # 5 due Friday
- Exam next week!

| Lec # | Date | | | Reading | HW |
|---|---|---|---|---|---|
| 12 | Mon | 9/30 | Leave one out CV | 5.1.1, 5.1.2 | |
| 13 | Wed | 10/2 | k-fold CV | 5.1.3 | |
| 14 | Fri | 10/4 | More k-fold CV, | 5.1.4-5 | |
| 15 | Mon | 10/7 | k-fold CV for classification | 5.1.5 | |
| 16 | Wed | 10/9 | Subset selection | 6.1 | HW #4 Due Weds 10/9 |
| 17 | Fri | 10/11 | Shrinkage: Ridge | 6.2.1 | |
| 18 | Mon | 10/14 | Shrinkage: Lasso | 6.2.2 | |
| 19 | Wed | 10/16 | Dimension Reduction PCA | 6.3 | |
| 20 | Fri | 10/18 | Overflow, Possibly more dimension reduction? PCR | | HW #5 Due Fri 10/18 |
| | Mon | 10/21 | No class - Fall break | | |
| | Wed | 10/23 | **Review** | | |
| | Fri | 10/25 | **Midterm #2** | | |

Section 1

# Last time

# Goal

- Fit model using all $p$ predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \underbrace{\beta_1}_{\to 0} X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

# Ridge regression

**Before:**

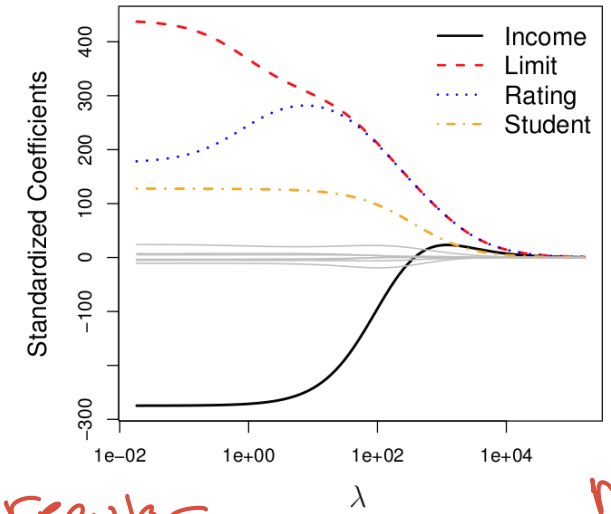
$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

**After:**

fixed $\lambda \geq 0$

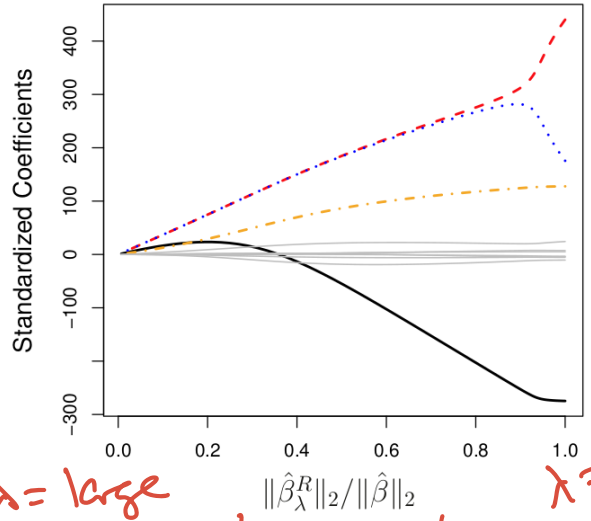
$(\|\beta_j\|_2)^2$

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$




regular linear regression $\lambda = 0$

null model large $\lambda$




$\lambda =$ large

how much shrinkage

$\lambda = 0$ ridge norm

lin reg norm

# Scale equivariance (or lack thereof)

**Scale equivariant:** Multiplying a variable by $c$ ($cX_i$) just returns a coefficient multiplied by $1/c$ ($1/c\beta_i$)

**Solution: standardize predictors**

$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

Std dev = 1

- Least squares is scale equivariant
- Ridge regression is not

Section 2

# The Lasso

# Same goal as before

- Fit model using all $p$ predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

Ridge regression shrunk the coef, but not actualy all the way to 0.

Today: Lasso, where coef end up actualy being 0!

"Least Absolute Shrinkage and Selection Operator"

# The lasso

**Least Squares:**

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

**Ridge:**

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

**The Lasso:**

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

$L_1$ penalty instead of a $L_2$ penalty

$$\|s\|_1 = \sum_{i=1}^{p}|s_i| \qquad \|s\|_2 = \sqrt{\sum_{i=1}^{p} s_i^2}$$

# Subsets with lasso

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

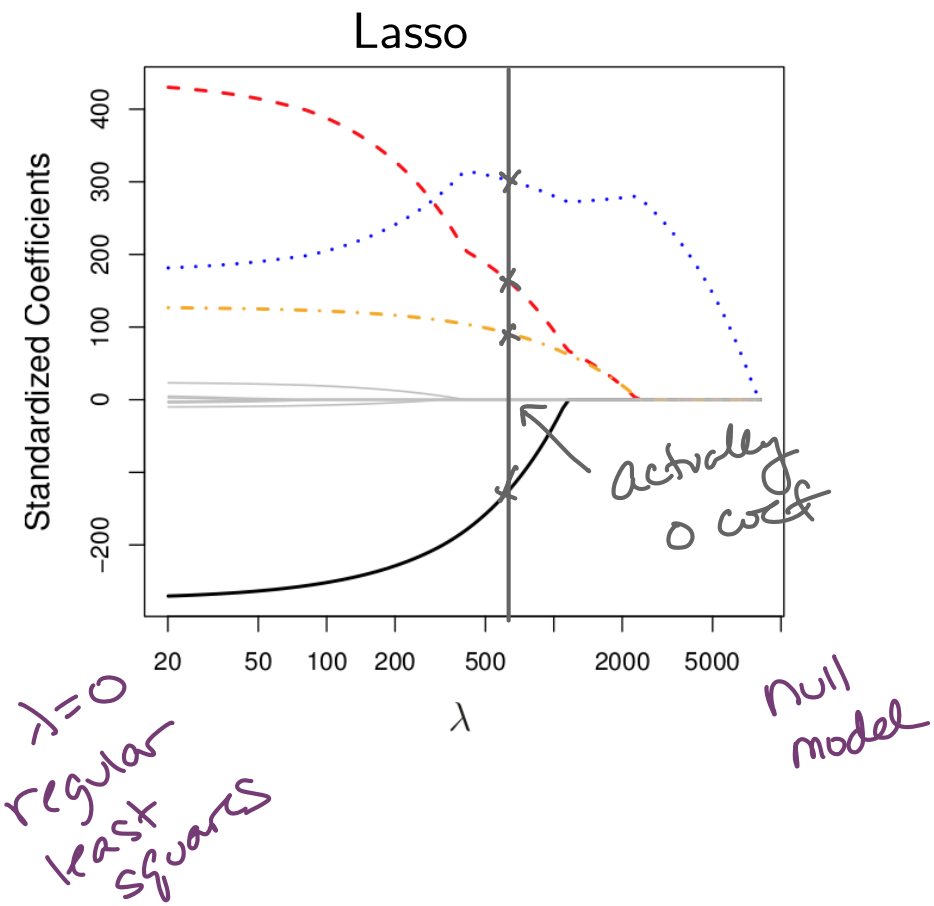
Result of $L_1$ penalty → forces some coef to be 0
(not just close)

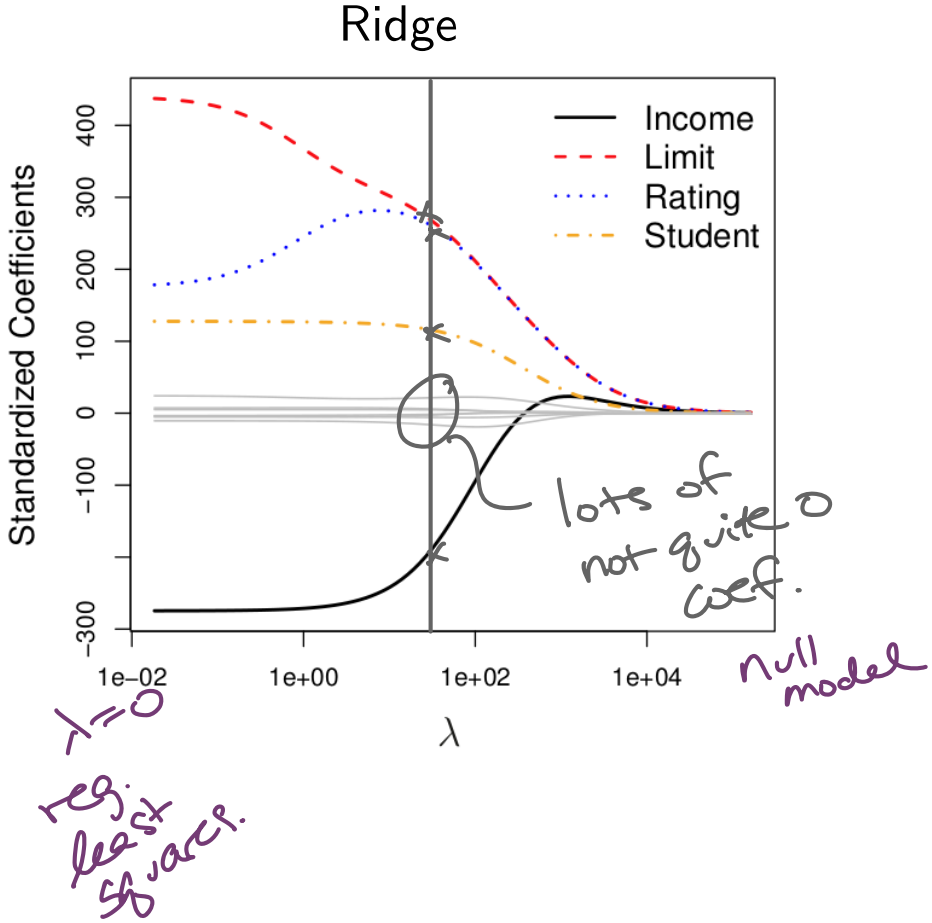
Can be interpreted as some kind subset selection,
instead here this is called variable selection

Result is a so-called sparse model
↳ lots of 0 coef

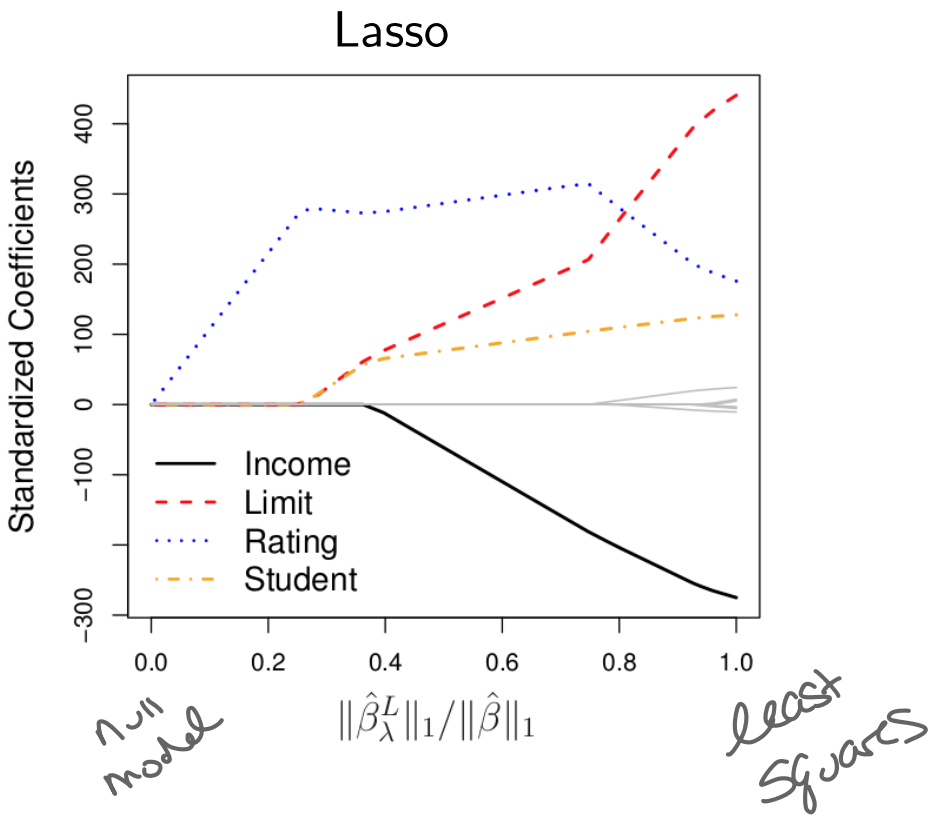
Goal: Need a good $\lambda$!

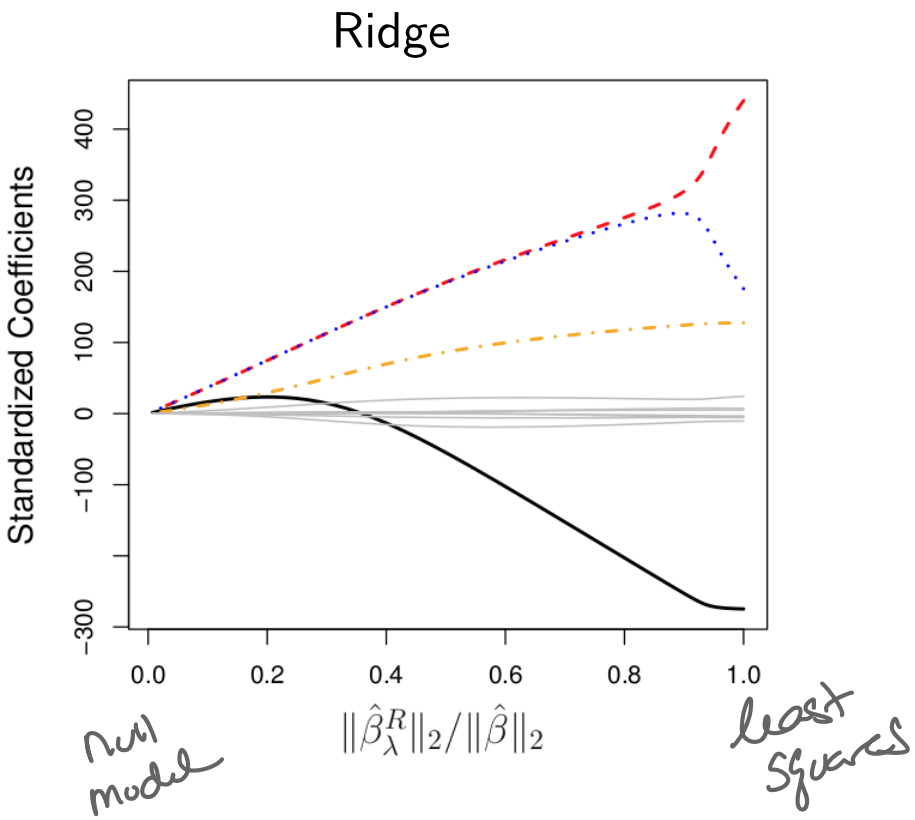
# An example on Credit data set



# More example on `Credit` data set



# Scale equivavariance (or lack thereof)

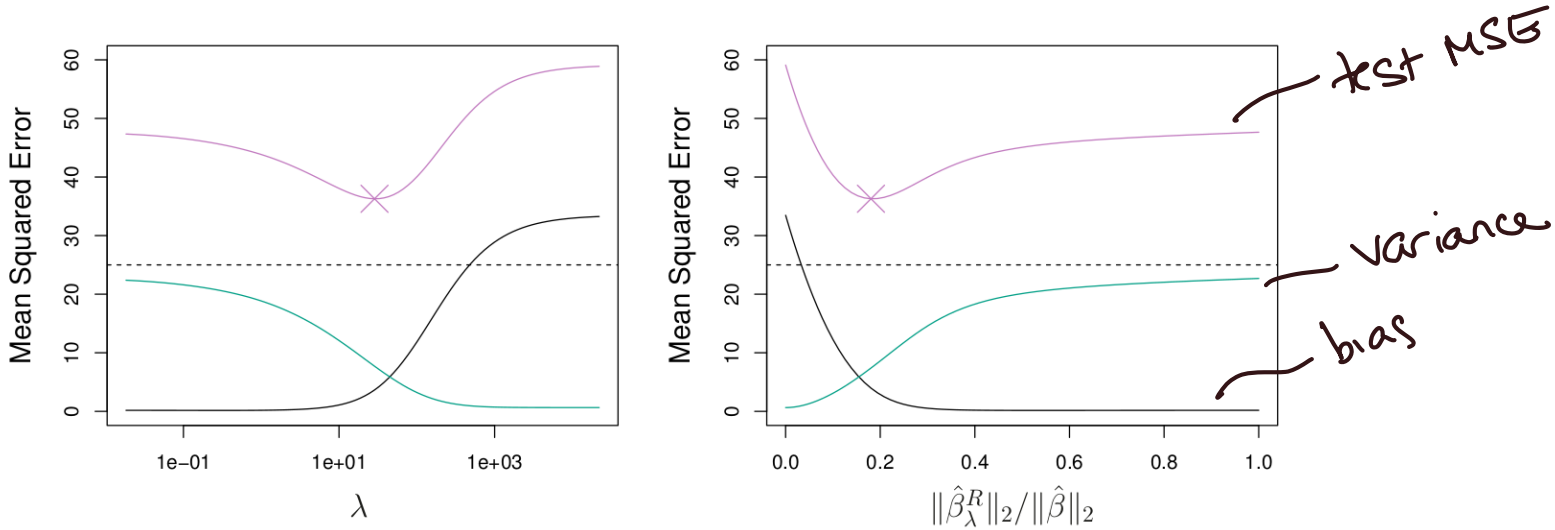
**Scale equivariant:** Multiplying a variable by $c$ just returns a coefficient multiplied by $1/c$

Least squares **is** scale equivariant.
Ridge/Lasso **are very much not**.

**Solution: standardize predictors**

$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

# Bias-Variance tradeoff

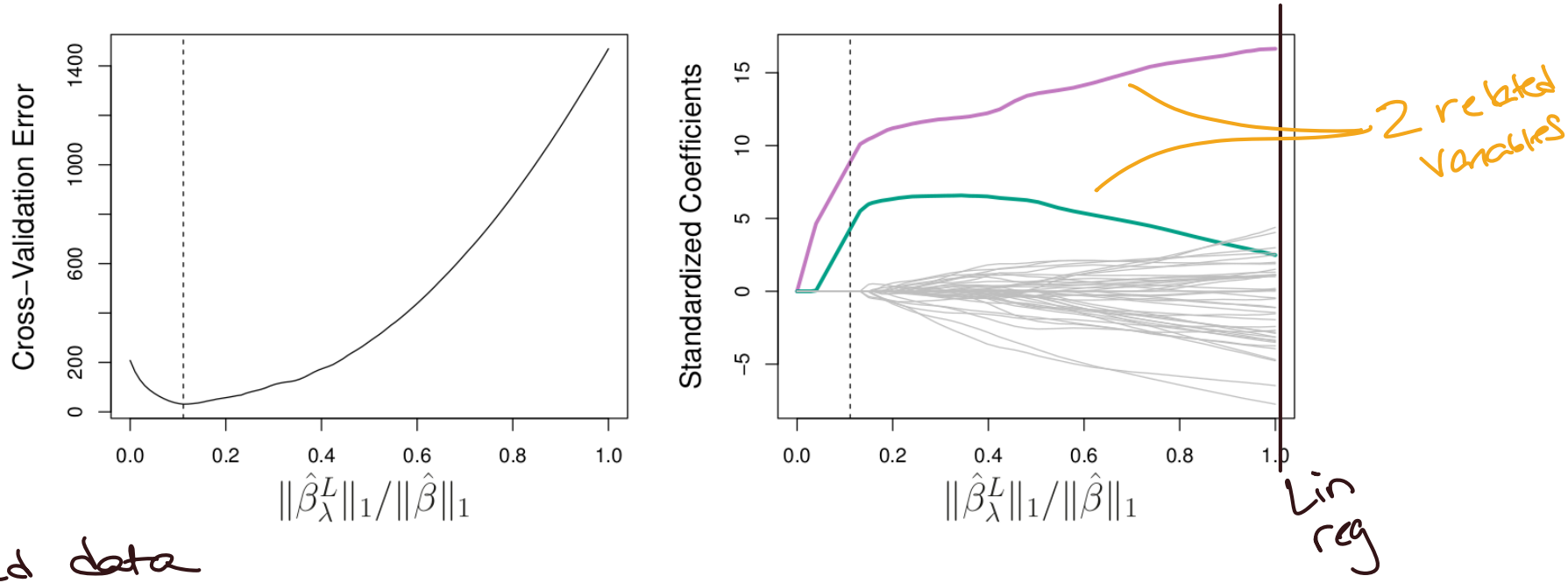



Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

# Using Cross-Validation to find $\lambda$

- Choose a grid of $\lambda$ values
- Compute the ($k$-fold) cross-validation error for each value of $\lambda$
- Select the tuning parameter value $\lambda$ for which the CV error is smallest.
- The model is re-fit using all of the available observations and the selected value of the tuning parameter.

# 10-fold CV choice of $\lambda$ for lasso and simulated data




Simulated data

p=45 variables

2 actually related to output Y

# Coding example

# Ridge vs Lasso

**Ridge Regression:**

Shrink coef to close to zero

Better than lasso if lots of relevant input variables

More robust to outliers

**Lasso:**

Force coef to actually be 0.

Better when only a few relevant variables used for output.

More interpretable (here are the coef that are non-zero)

# TL;DR - Original forumlation

**Least Squares:**

$$RSS = \sum_{i=1}^{n} \left( y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

**Ridge:**

$$\sum_{i=1}^{n} \left( y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 = RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$

**The Lasso:**

$$\sum_{i=1}^{n} \left( y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| = RSS + \lambda \sum_{j=1}^{p} |\beta_j|$$

# Next time

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